1) a)

$\begin{pmatrix}4 & 1\\ 2 & 3\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = \lambda\begin{pmatrix}x\\ y\end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix}4 & 1\\ 2 & 3\end{pmatrix} - \lambda\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = \begin{pmatrix}4-\lambda & 1\\ 2 & 3-\lambda\end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10$

So $(\lambda - 2)(\lambda - 5) = 0$

$\therefore \lambda = 2$ or $\lambda = 5$

When $\lambda_1 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix}2 & 1\\ 2 & 1\end{pmatrix}$

$\begin{pmatrix}2 & 1\\ 2 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = 0 \Rightarrow 2x + y = 0$ so $y = -2x \therefore \mathbf{v}_1 = \begin{pmatrix}1\\ -2\end{pmatrix}$

When $\lambda_2 = 5$ then $\mathbf{A} - 5\mathbf{I} = \begin{pmatrix}-1 & 1\\ 2 & -2\end{pmatrix}$

$\begin{pmatrix}-1 & 1\\ 2 & -2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = 0 \Rightarrow -x + y = 0$ so $y = x \therefore \mathbf{v}_2 = \begin{pmatrix}1\\ 1\end{pmatrix}$

$\therefore \mathbf{P} = \begin{pmatrix}1 & 1\\ -2 & 1\end{pmatrix}$

1) b)

$\mathbf{P}^{-1} = \frac{1}{3}\begin{pmatrix}1 & -1\\ 2 & 1\end{pmatrix}$

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$

$= \frac{1}{3}\begin{pmatrix}1 & -1\\ 2 & 1\end{pmatrix}\begin{pmatrix}4 & 1\\ 2 & 3\end{pmatrix}\begin{pmatrix}1 & 1\\ -2 & 1\end{pmatrix} = \frac{1}{3}\begin{pmatrix}2 & -2\\ 10 & 5\end{pmatrix}\begin{pmatrix}1 & 1\\ -2 & 1\end{pmatrix} = \frac{1}{3}\begin{pmatrix}6 & 0\\ 0 & 15\end{pmatrix} = \begin{pmatrix}2 & 0\\ 0 & 5\end{pmatrix}$

$\therefore \mathbf{D} = \begin{pmatrix}2 & 0\\ 0 & 5\end{pmatrix}$

2) a)

$\begin{pmatrix}7 & 2\\ 3 & 6\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\end{pmatrix}$

$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}7 & 2\\ 3 & 6\end{pmatrix}-\lambda\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}=\begin{pmatrix}7-\lambda & 2\\ 3 & 6-\lambda\end{pmatrix}$

$\det(\mathbf{A}-\lambda\mathbf{I})=(7-\lambda)(6-\lambda)-6=\lambda^2-13\lambda+36$

So $(\lambda-4)(\lambda-9)=0$

$\therefore \lambda=4$ or $\lambda=9$

When $\lambda_1=4$ then $\mathbf{A}-4\mathbf{I}=\begin{pmatrix}3 & 2\\ 3 & 2\end{pmatrix}$

$\begin{pmatrix}3 & 2\\ 3 & 2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=0\Rightarrow 3x+2y=0$ so $2y=-3x\therefore \mathbf{v}_1=\begin{pmatrix}2\\ -3\end{pmatrix}$

When $\lambda_2=9$ then $\mathbf{A}-9\mathbf{I}=\begin{pmatrix}-2 & 2\\ 3 & -3\end{pmatrix}$

$\begin{pmatrix}-2 & 2\\ 3 & -3\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=0\Rightarrow -2x+2y=0$ so $x=y\therefore \mathbf{v}_2=\begin{pmatrix}1\\ 1\end{pmatrix}$

$\therefore \mathbf{P}=\begin{pmatrix}2 & 1\\ -3 & 1\end{pmatrix}$

$\mathbf{P}^{-1}=\frac{1}{5}\begin{pmatrix}1 & -1\\ 3 & 2\end{pmatrix}=\begin{pmatrix}\frac{1}{5} & -\frac{1}{5}\\ \frac{3}{5} & \frac{2}{5}\end{pmatrix}$

$\mathbf{D}=\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$

$=\begin{pmatrix}\frac{1}{5} & -\frac{1}{5}\\ \frac{3}{5} & \frac{2}{5}\end{pmatrix}\begin{pmatrix}7 & 2\\ 3 & 6\end{pmatrix}\begin{pmatrix}2 & 1\\ -3 & 1\end{pmatrix}=\begin{pmatrix}\frac{1}{5} & -\frac{1}{5}\\ \frac{3}{5} & \frac{2}{5}\end{pmatrix}\begin{pmatrix}8 & 9\\ -12 & 9\end{pmatrix}=\begin{pmatrix}4 & 0\\ 0 & 9\end{pmatrix}$

$\therefore \mathbf{P}=\begin{pmatrix}2 & 1\\ -3 & 1\end{pmatrix}$ and $\mathbf{D}=\begin{pmatrix}4 & 0\\ 0 & 9\end{pmatrix}$

2) b)

$\begin{pmatrix}20 & 12\\15 & 17\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\lambda\begin{pmatrix}x\\y\end{pmatrix}$

$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}20 & 12\\15 & 17\end{pmatrix}-\lambda\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}=\begin{pmatrix}20-\lambda & 12\\15 & 17-\lambda\end{pmatrix}$

$\det(\mathbf{A}-\lambda\mathbf{I})=(20-\lambda)(17-\lambda)-180=\lambda^2-37\lambda+160$

So $(\lambda-5)(\lambda-32)=0$

$\therefore \lambda=5$ or $\lambda=32$

When $\lambda_1=4$ then $\mathbf{A}-5\mathbf{I}=\begin{pmatrix}15 & 12\\15 & 12\end{pmatrix}$

$\begin{pmatrix}15 & 12\\15 & 12\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=0\Rightarrow 15x+12y=0$ so $5x=-4y\therefore \mathbf{v}_1=\begin{pmatrix}4\\-5\end{pmatrix}$

When $\lambda_2=32$ then $\mathbf{A}-32\mathbf{I}=\begin{pmatrix}-12 & 12\\15 & -15\end{pmatrix}$

$\begin{pmatrix}-12 & 12\\15 & -15\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=0\Rightarrow -12x+12y=0$ so $x=y\therefore \mathbf{v}_2=\begin{pmatrix}1\\1\end{pmatrix}$

$\therefore \mathbf{P}=\begin{pmatrix}4 & 1\\-5 & 1\end{pmatrix}$

$\mathbf{P}^{-1}=\frac{1}{9}\begin{pmatrix}1 & -1\\5 & 4\end{pmatrix}$

$\mathbf{D}=\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$

$=\frac{1}{9}\begin{pmatrix}1 & -1\\5 & 4\end{pmatrix}\begin{pmatrix}20 & 12\\15 & 17\end{pmatrix}\begin{pmatrix}4 & 1\\-5 & 1\end{pmatrix}=\frac{1}{9}\begin{pmatrix}1 & -1\\5 & 4\end{pmatrix}\begin{pmatrix}20 & 32\\-25 & 32\end{pmatrix}=\frac{1}{9}\begin{pmatrix}45 & 0\\0 & 288\end{pmatrix}=\begin{pmatrix}5 & 0\\0 & 32\end{pmatrix}$

$\therefore \mathbf{P}=\begin{pmatrix}4 & 1\\-5 & 1\end{pmatrix}$ and $\mathbf{D}=\begin{pmatrix}5 & 0\\0 & 32\end{pmatrix}$

2) c)

$\begin{pmatrix} 12 & 7 \\ -14 & -9 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 12 & 7 \\ -14 & -9 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 12-\lambda & 7 \\ -14 & -9-\lambda \end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (12-\lambda)(-9-\lambda) + 98 = \lambda^2 - 3\lambda - 10$

So $(\lambda + 2)(\lambda - 5) = 0$

$\therefore \lambda = -2$ or $\lambda = 5$

When $\lambda_1 = -2$ then $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 14 & 7 \\ -14 & -7 \end{pmatrix}$

$\begin{pmatrix} 14 & 7 \\ -14 & -7 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow 14x + 7y = 0$ so $y = -2x \therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

When $\lambda_2 = 5$ then $\mathbf{A} - 5\mathbf{I} = \begin{pmatrix} 7 & 7 \\ -14 & -14 \end{pmatrix}$

$\begin{pmatrix} 7 & 7 \\ -14 & -14 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow x + y = 0$ so $x = y \therefore \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\therefore \mathbf{P} = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix}$

$\mathbf{P}^{-1} = \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}$

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$

$= \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 12 & 7 \\ -14 & -9 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} -2 & 5 \\ 4 & -5 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & 5 \end{pmatrix}$

$\therefore \mathbf{P} = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} -2 & 0 \\ 0 & 5 \end{pmatrix}$

3) a)

$\begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 5-\lambda & 1 \\ 0 & 5-\lambda \end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (5-\lambda)(5-\lambda)$

So $(5-\lambda)(5-\lambda) = 0$

$\therefore \lambda = 5$ or $\lambda = 5$

When $\lambda = 5$ then $\mathbf{A} - 5\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow y = 0$ so $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

3) b)

A matrix is only diagonalisable if it has enough linearly independent eigenvectors and in this case there is only one independent eigenvector.

4) a)

$\begin{pmatrix}13 & 6\\ 3 & 10\end{pmatrix}\begin{pmatrix}1\\ -1\end{pmatrix} = \begin{pmatrix}7\\ -7\end{pmatrix} = 7\begin{pmatrix}1\\ -1\end{pmatrix}$ So $\lambda = 7$ has eigenvector $\begin{pmatrix}1\\ -1\end{pmatrix}$

$\begin{pmatrix}13 & 6\\ 3 & 10\end{pmatrix}\begin{pmatrix}2\\ 1\end{pmatrix} = \begin{pmatrix}32\\ 16\end{pmatrix} = 16\begin{pmatrix}2\\ 1\end{pmatrix}$ So $\lambda = 16$ has eigenvector $\begin{pmatrix}2\\ 1\end{pmatrix}$

He incorrectly constructs matrices $\mathbf{P}$ and $\mathbf{D}$; he should have used $\begin{pmatrix}7 & 0\\ 0 & 16\end{pmatrix}\begin{pmatrix}1 & 2\\ -1 & 1\end{pmatrix}$ or $\begin{pmatrix}16 & 0\\ 0 & 7\end{pmatrix}\begin{pmatrix}2 & 1\\ 1 & -1\end{pmatrix}$

When computing $\mathbf{P}^{-1}$ he did not divide by the determinant.

4) b)

$\mathbf{P} = \begin{pmatrix}1 & 2\\ -1 & 1\end{pmatrix}$

So $\frac{1}{3}\begin{pmatrix}1 & 2\\ -1 & 1\end{pmatrix}\begin{pmatrix}7 & 0\\ 0 & 16\end{pmatrix}\begin{pmatrix}1 & -2\\ 1 & 1\end{pmatrix} = \frac{1}{3}\begin{pmatrix}1 & 2\\ -1 & 1\end{pmatrix}\begin{pmatrix}7 & -14\\ 16 & 16\end{pmatrix} = \frac{1}{3}\begin{pmatrix}39 & 18\\ 9 & 30\end{pmatrix} = \begin{pmatrix}13 & 6\\ 3 & 10\end{pmatrix}$

5) a)

$\begin{pmatrix}5 & 0\\ 0 & 4\end{pmatrix}^8 = \begin{pmatrix}5^8 & 0\\ 0 & 4^8\end{pmatrix} = \begin{pmatrix}390625 & 0\\ 0 & 65536\end{pmatrix}$

5) b)

$\begin{pmatrix}1 & 0\\ 0 & -2\end{pmatrix}^{10} = \begin{pmatrix}1^{10} & 0\\ 0 & (-2)^{10}\end{pmatrix} = \begin{pmatrix}1 & 0\\ 0 & 1024\end{pmatrix}$

5) c)

$\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}^k = \begin{pmatrix}a^k & 0\\ 0 & b^k\end{pmatrix}$

6)

$\mathbf{M}^n = (\mathbf{PDP}^{-1})^n$

$= (\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1}) \ldots (\mathbf{PDP}^{-1})$

$= \mathbf{PD}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1} \ldots \mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P})\mathbf{DP}^{-1}$

$= \mathbf{PDDD} \ldots \mathbf{DDDP}^{-1}$

$= \mathbf{PD}^{\boldsymbol{n}}\mathbf{P}^{-1}$ as req.

7) a)

$\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3-\lambda & 1 \\ 0 & 2-\lambda \end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (3-\lambda)(2-\lambda)$

So $(3-\lambda)(2-\lambda) = 0$

$\therefore \lambda = 2$ or $\lambda = 3$

When $\lambda_1 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow x + y = 0$ so $y = -x \therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

When $\lambda_2 = 3$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix}$

$\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow y = 0$ and x can take any value $\therefore \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\therefore\ \lambda_1 = 2, \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\lambda_2 = 3, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

7) b)

$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$

7) c)

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP} \Rightarrow \mathbf{PD} = \mathbf{AP} \Rightarrow \mathbf{PDP}^{-1} = \mathbf{A} \therefore \mathbf{A}^4 = \mathbf{PD}^4\mathbf{P}^{-1}$

$\mathbf{P}^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$

$\mathbf{A}^4 = \mathbf{PD}^4\mathbf{P}^{-1} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}^{\mathbf{4}}\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 16 & 0 \\ 0 & 81 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -16 \\ 81 & 81 \end{pmatrix}$

$= \begin{pmatrix} 81 & 65 \\ 0 & 16 \end{pmatrix}$

8) a)

$\begin{pmatrix} -4 & 0 \\ -15 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} -4 & 0 \\ -15 & 1 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -4-\lambda & 0 \\ -15 & 1-\lambda \end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (-4-\lambda)(1-\lambda)$

So $(-4-\lambda)(1-\lambda) = 0$

$\therefore \lambda = -4$ or $\lambda = 1$

8) b)

When $\lambda_1 = -4$ then $\mathbf{A} + 4\mathbf{I} = \begin{pmatrix} 0 & 0 \\ -15 & 5 \end{pmatrix}$

$\begin{pmatrix} 0 & 0 \\ -15 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -15x + 5y = 0$ so $y = 3x \therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

When $\lambda_2 = 1$ then $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -5 & 0 \\ -15 & 0 \end{pmatrix}$

$\begin{pmatrix} -5 & 0 \\ -15 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -5x = 0 \therefore x = 0$ and y can take any value $\therefore \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

8) c)

$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} -4 & 0 \\ 0 & 1 \end{pmatrix}$

8) d)

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP} \Rightarrow \mathbf{PD} = \mathbf{AP} \Rightarrow \mathbf{PDP}^{-1} = \mathbf{A} \therefore \mathbf{A}^4 = \mathbf{PD}^4\mathbf{P}^{-1}$

$\mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}$

$\mathbf{A}^{50} = \mathbf{PD}^{50}\mathbf{P}^{-1} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} -4 & 0 \\ 0 & 1 \end{pmatrix}^{50}\begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} 2^{100} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} 2^{100} & 0 \\ -3 & 1 \end{pmatrix}$

$= \begin{pmatrix} 2^{100} & 0 \\ 3 \times 2^{100} - 3 & 1 \end{pmatrix}$

9) a)

$\mathbf{P} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix}$

$\mathbf{M}^n = \mathbf{PD}^n\mathbf{P}^{-1}$

$\mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$

$\mathbf{M}^n = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix}^n\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4^n} \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{4^n} \end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix} 2 & 1 - \frac{1}{4^n} \\ 0 & \frac{2}{4^n} \end{pmatrix}$

$= \begin{pmatrix} 1 & \frac{1}{2}\left(1 - \frac{1}{4^n}\right) \\ 0 & \frac{1}{4^n} \end{pmatrix}$

9) b)

$\mathbf{L} = \lim_{n \to \infty} \begin{pmatrix} 1 & \frac{1}{2}\left(1 - \frac{1}{4^n}\right) \\ 0 & \frac{1}{4^n} \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{pmatrix}$

10) a)

$$\begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{6} & \frac{5}{6} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{6} & \frac{5}{6} \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} - \lambda & \frac{1}{3} \\ \frac{1}{6} & \frac{5}{6} - \lambda \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \left(\frac{2}{3} - \lambda\right)\left(\frac{5}{6} - \lambda\right) - \frac{1}{18} = \lambda^2 - \frac{3}{2}\lambda + \frac{1}{2}$$

So $\frac{1}{2}(2\lambda - 1)(\lambda - 1) = 0$

$\therefore \lambda = \frac{1}{2}$ or $\lambda = 1$

When $\lambda_1 = \frac{1}{2}$ then $\mathbf{A} - \frac{1}{2}\mathbf{I} = \begin{pmatrix} \frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} \end{pmatrix}$

$$\begin{pmatrix} \frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow \frac{1}{6}x + \frac{1}{3}y = 0 \text{ so } y = -\frac{1}{2}x \therefore \mathbf{v}_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

When $\lambda_2 = 1$ then $\mathbf{A} - \mathbf{I} = \begin{pmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & -\frac{1}{6} \end{pmatrix}$

$$\begin{pmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{6} & -\frac{1}{6} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -\frac{1}{3}x + \frac{1}{3}y = 0 \therefore y = x \therefore \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$\mathbf{P} = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$

$\mathbf{M}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$

$\mathbf{P}^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$

$$\mathbf{M}^n = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}^n\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{2^n} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{2^n} & -\frac{1}{2^n} \\ 1 & 2 \end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix} \frac{2}{2^n} + 1 & 2 - \frac{2}{2^n} \\ 1 - \frac{1}{2^n} & 2 + \frac{1}{2^n} \end{pmatrix}$$

10) b)

$$\mathbf{L} = \lim_{n\to\infty}\left(\frac{1}{3}\begin{pmatrix}\frac{2}{2^n}+1 & 2-\frac{2}{2^n}\\ 1-\frac{1}{2^n} & 2+\frac{1}{2^n}\end{pmatrix}\right) = \frac{1}{3}\begin{pmatrix}1 & 2\\ 1 & 2\end{pmatrix}$$

11)

$$\begin{pmatrix}4 & 3\\ 0 & 2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = \lambda\begin{pmatrix}x\\ y\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I} = \begin{pmatrix}4 & 3\\ 0 & 2\end{pmatrix} - \lambda\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = \begin{pmatrix}4-\lambda & 3\\ 0 & 2-\lambda\end{pmatrix}$$

$\det(\mathbf{A}-\lambda\mathbf{I}) = (4-\lambda)(2-\lambda)$

So $(4-\lambda)(2-\lambda) = 0$

$\therefore \lambda = 2$ or $\lambda = 4$

When $\lambda_1 = 2$ then $\mathbf{A}-2\mathbf{I} = \begin{pmatrix}2 & 3\\ 0 & 0\end{pmatrix}$

$\begin{pmatrix}2 & 3\\ 0 & 0\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = 0 \Rightarrow 2x+3y=0$ so $3y=-2x \therefore \mathbf{v}_1 = \begin{pmatrix}3\\ -2\end{pmatrix}$

When $\lambda_2 = 4$ then $\mathbf{A}-4\mathbf{I} = \begin{pmatrix}0 & 3\\ 0 & -2\end{pmatrix}$

$\begin{pmatrix}0 & 3\\ 0 & -2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = 0 \Rightarrow 3y=0 \therefore y=0$ and x can take any value $\therefore \mathbf{v}_2 = \begin{pmatrix}1\\ 0\end{pmatrix}$

$\mathbf{P} = \begin{pmatrix}3 & 1\\ -2 & 0\end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix}2 & 0\\ 0 & 4\end{pmatrix}$

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} \Rightarrow \mathbf{P}\mathbf{D} = \mathbf{A}\mathbf{P} \Rightarrow \mathbf{P}\mathbf{D}\mathbf{P}^{-1} = \mathbf{A} \therefore \mathbf{A}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$

$\mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix}0 & -1\\ 2 & 3\end{pmatrix}$

$$\mathbf{A}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} = \frac{1}{2}\begin{pmatrix}3 & 1\\ -2 & 0\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & 4\end{pmatrix}^n\begin{pmatrix}0 & -1\\ 2 & 3\end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix}3 & 1\\ -2 & 0\end{pmatrix}\begin{pmatrix}2^n & 0\\ 0 & 4^n\end{pmatrix}\begin{pmatrix}0 & -1\\ 2 & 3\end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix}3 & 1\\ -2 & 0\end{pmatrix}\begin{pmatrix}0 & -2^n\\ 2\times 4^n & 3\times 4^n\end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix}2\times 4^n & 3\times 4^n - 3\times 2^n\\ 0 & 2\times 2^n\end{pmatrix}$$

$$= \begin{pmatrix}4^n & \frac{3}{2}(4^n-2^n)\\ 0 & 2^n\end{pmatrix}$$ as req.

12)

$\begin{pmatrix}2 & 6\\ 1 & 3\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\end{pmatrix}$

$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}2 & 6\\ 1 & 3\end{pmatrix}-\lambda\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}=\begin{pmatrix}2-\lambda & 6\\ 1 & 3-\lambda\end{pmatrix}$

$\det(\mathbf{A}-\lambda\mathbf{I})=(2-\lambda)(3-\lambda)-6=\lambda^2-5\lambda$

So $\lambda(\lambda-5)=0$

$\therefore \lambda=0$ or $\lambda=5$

When $\lambda_1=0$ then $\mathbf{A}-0\mathbf{I}=\begin{pmatrix}2 & 6\\ 1 & 3\end{pmatrix}$

$\begin{pmatrix}2 & 6\\ 1 & 3\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=0\Rightarrow x+3y=0$ so $-3y=x\therefore \mathbf{v}_1=\begin{pmatrix}-3\\ 1\end{pmatrix}$

When $\lambda_2=5$ then $\mathbf{A}-5\mathbf{I}=\begin{pmatrix}-3 & 6\\ 1 & -2\end{pmatrix}$

$\begin{pmatrix}-3 & 6\\ 1 & -2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=0\Rightarrow x-2y=0\therefore x=2y\therefore \mathbf{v}_2=\begin{pmatrix}2\\ 1\end{pmatrix}$

$\mathbf{P}=\begin{pmatrix}-3 & 2\\ 1 & 1\end{pmatrix}$ and $\mathbf{D}=\begin{pmatrix}0 & 0\\ 0 & 5\end{pmatrix}$

$\mathbf{D}=\mathbf{P}^{-1}\mathbf{A}\mathbf{P}\Rightarrow\mathbf{P}\mathbf{D}=\mathbf{A}\mathbf{P}\Rightarrow\mathbf{P}\mathbf{D}\mathbf{P}^{-1}=\mathbf{A}\therefore\mathbf{A}^n=\mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$

$\mathbf{P}^{-1}=-\frac{1}{5}\begin{pmatrix}1 & -2\\ -1 & -3\end{pmatrix}$

$\mathbf{A}^n=\mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}=-\frac{1}{5}\begin{pmatrix}-3 & 2\\ 1 & 1\end{pmatrix}\begin{pmatrix}0 & 0\\ 0 & 5\end{pmatrix}^n\begin{pmatrix}1 & -2\\ -1 & -3\end{pmatrix}$

$=-\frac{1}{5}\begin{pmatrix}-3 & 2\\ 1 & 1\end{pmatrix}\begin{pmatrix}0 & 0\\ 0 & 5^n\end{pmatrix}\begin{pmatrix}1 & -2\\ -1 & -3\end{pmatrix}$

$=-\frac{1}{5}\begin{pmatrix}-3 & 2\\ 1 & 1\end{pmatrix}\begin{pmatrix}0 & 0\\ -5^n & -3\times5^n\end{pmatrix}$

$=-\frac{1}{5}\begin{pmatrix}-2\times5^n & -6\times5^n\\ -5^n & -3\times5^n\end{pmatrix}$

$=\frac{1}{5}\begin{pmatrix}2\times5^n & 6\times5^n\\ 5^n & 3\times5^n\end{pmatrix}$

$=\begin{pmatrix}2\times5^{n-1} & 6\times5^{n-1}\\ 5^{n-1} & 3\times5^{n-1}\end{pmatrix}$

$=5^{n-1}\begin{pmatrix}2 & 6\\ 1 & 3\end{pmatrix}$ as req.

13)

$\begin{pmatrix} p & q \\ p & q \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} p & q \\ p & q \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p-\lambda & q \\ p & q-\lambda \end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (p-\lambda)(q-\lambda) - pq = \lambda^2 - \lambda(p+q)$

So $\lambda\big(\lambda - (p+q)\big) = 0$

$\therefore \lambda = 0$ or $\lambda = p+q$

When $\lambda_1 = -k$ then $\mathbf{A} + 0\mathbf{I} = \begin{pmatrix} p & q \\ p & q \end{pmatrix}$

$\begin{pmatrix} p & q \\ p & q \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow px + qy = 0$ so $qy = -px \therefore \mathbf{v}_1 = \begin{pmatrix} q \\ -p \end{pmatrix}$

When $\lambda_2 = p+q$ then $\mathbf{A} - (p+q)\mathbf{I} = \begin{pmatrix} -q & q \\ p & -q \end{pmatrix}$

$\begin{pmatrix} -q & q \\ p & -q \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow -qx + qy = 0 \therefore y = x \therefore \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\mathbf{P} = \begin{pmatrix} q & 1 \\ -p & 1 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & p+q \end{pmatrix}$

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} \Rightarrow \mathbf{P}\mathbf{D} = \mathbf{A}\mathbf{P} \Rightarrow \mathbf{P}\mathbf{D}\mathbf{P}^{-1} = \mathbf{A} \therefore \mathbf{A}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$

$\mathbf{P}^{-1} = \frac{1}{p-q}\begin{pmatrix} 1 & -1 \\ p & q \end{pmatrix}$

$\mathbf{A}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1} = \frac{1}{p-q}\begin{pmatrix} q & 1 \\ -p & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & p+q \end{pmatrix}^n\begin{pmatrix} 1 & -1 \\ p & q \end{pmatrix}$

$= \frac{1}{p+q}\begin{pmatrix} q & 1 \\ -p & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & (p+q)^n \end{pmatrix}\begin{pmatrix} 1 & -1 \\ p & q \end{pmatrix}$

$= \frac{1}{p+q}\begin{pmatrix} q & 1 \\ -p & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ p(p+q)^n & q(p+q)^n \end{pmatrix}$

$= \frac{1}{p+q}\begin{pmatrix} p(p+q)^n & q(p+q)^n \\ p(p+q)^n & q(p+q)^n \end{pmatrix}$

$= \frac{(p+q)^n}{p+q}\begin{pmatrix} p & q \\ p & q \end{pmatrix}$

$= (p+q)^{n-1}\begin{pmatrix} p & q \\ p & q \end{pmatrix} = (p+q)^{n-1}\mathbf{A}$ as req.

14)

$\begin{pmatrix} 1 & k+1 \\ k-1 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$

$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 1 & k+1 \\ k-1 & -1 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1-\lambda & k+1 \\ k-1 & -1-\lambda \end{pmatrix}$

$\det(\mathbf{A} - \lambda\mathbf{I}) = (1-\lambda)(-1-\lambda) - (k+1)(k-1) = \lambda^2 - 1 + 1 - k^2 = \lambda^2 - k^2$

So $(\lambda + k)(\lambda - k) = 0$

$\therefore \lambda = -k$ or $\lambda = k$

When $\lambda_1 = -k$ then $\mathbf{A} + k\mathbf{I} = \begin{pmatrix} 1+k & k+1 \\ k-1 & -1+k \end{pmatrix}$

$\begin{pmatrix} 1+k & k+1 \\ k-1 & -1+k \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow (k+1)x + (k+1)y = 0$ so $x = -y \therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

When $\lambda_2 = k$ then $\mathbf{A} - k\mathbf{I} = \begin{pmatrix} 1-k & k+1 \\ k-1 & -1-k \end{pmatrix}$

$\begin{pmatrix} 1-k & k+1 \\ k-1 & -1-k \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow (1-k)x + (k+1)y = 0 \therefore y = \frac{k-1}{k+1}x \therefore \mathbf{v}_2 = \begin{pmatrix} k+1 \\ k-1 \end{pmatrix}$

$\mathbf{P} = \begin{pmatrix} 1 & k+1 \\ -1 & k-1 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} -k & 0 \\ 0 & k \end{pmatrix}$

$\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} \Rightarrow \mathbf{P}\mathbf{D} = \mathbf{A}\mathbf{P} \Rightarrow \mathbf{P}\mathbf{D}\mathbf{P}^{-1} = \mathbf{A} \therefore \mathbf{A}^n = \mathbf{P}\mathbf{D}^n\mathbf{P}^{-1}$

$\mathbf{P}^{-1} = \frac{1}{2k}\begin{pmatrix} k-1 & -k-1 \\ 1 & 1 \end{pmatrix}$

$\mathbf{A}^{2n+1} = \mathbf{P}\mathbf{D}^{2n+1}\mathbf{P}^{-1} = \frac{1}{2k}\begin{pmatrix} 1 & k+1 \\ -1 & k-1 \end{pmatrix}\begin{pmatrix} -k & 0 \\ 0 & k \end{pmatrix}^{2n+1}\begin{pmatrix} k-1 & -k-1 \\ 1 & 1 \end{pmatrix}$

$= \frac{1}{2k}\begin{pmatrix} 1 & k+1 \\ -1 & k-1 \end{pmatrix}\begin{pmatrix} -k^{2n+1} & 0 \\ 0 & k^{2n+1} \end{pmatrix}\begin{pmatrix} k-1 & -k-1 \\ 1 & 1 \end{pmatrix}$

$= \frac{1}{2k}\begin{pmatrix} 1 & k+1 \\ -1 & k-1 \end{pmatrix}\begin{pmatrix} -(k-1)k^{2n+1} & (k+1)k^{2n+1} \\ k^{2n+1} & k^{2n+1} \end{pmatrix}$

$= \frac{1}{2}k^{2n}\begin{pmatrix} 1 & k+1 \\ -1 & k-1 \end{pmatrix}\begin{pmatrix} -k+1 & k+1 \\ 1 & 1 \end{pmatrix}$

$= \frac{1}{2}k^{2n}\begin{pmatrix} 2 & 2(k+1) \\ 2(k-1) & -2 \end{pmatrix}$

$= k^{2n}\begin{pmatrix} 1 & k+1 \\ k-1 & -1 \end{pmatrix} = k^{2n}\mathbf{A}^n$ as req.